Banking Customer Analytics
Detailed Project Documentation

1. Business Objective

The objective of this project was to analyze banking customer, account, loan, transaction, card, merchant, and branch data to understand customer profiles, financial activity, loan exposure, and banking product usage.

The project uses **MySQL for data analysis** and **Power BI for interactive visualization and dashboard development**.

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2. Dataset

The dataset contains seven main tables:

- **Customers** – customer information and credit scores
- **Accounts** – account types, balances, and opening dates
- **Loans** – loan amounts, interest rates, and start dates
- **Transactions** – transaction amounts, dates, accounts, and merchants
- **Cards** – card types and expiration dates
- **Merchants** – merchant names and locations
- **Branches** – branch names, managers, cities, and countries

The tables were analyzed individually and through relationships between customers, accounts, loans, transactions, and merchants.

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3. Tools & Technologies

- **MySQL** – Data analysis and SQL queries
- **Microsoft Power BI** – Dashboard and visualization
- **DAX** – Measures and calculated columns
- **Power Query** – Data preparation

- **GitHub** – Project documentation and portfolio

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4. Data Preparation & Analysis

The analysis included:

- Data structure and quality checks
- Customer credit score analysis
- Account balance analysis
- Account type analysis
- Loan portfolio analysis
- Interest rate analysis
- Transaction trend analysis
- Customer city analysis
- Merchant transaction analysis
- Card type and expiration analysis
- Customer account ownership analysis

Customers were grouped into five credit score categories:

Credit Score	Category
Below 580	Poor
580–669	Fair
670–739	Good
740–799	Very Good
800+	Excellent

Loan interest rates were also grouped into four categories to simplify analysis.

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5. Key Performance Indicators

The Power BI dashboard includes the following KPIs:

KPI	Purpose
Total Customers	Measures customer base
Total Accounts	Measures account volume
Total Account Balance	Measures funds held in accounts
Total Loans	Measures loan volume
Total Loan Value	Measures loan exposure
Total Cards	Measures card volume
Total Transactions	Measures transaction volume
Total Transaction Value	Measures transaction activity
Average Credit Score	Measures overall customer credit profile

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6. Power BI Dashboard

The Power BI report contains **three pages**.

Executive Overview

Provides a high-level view of the banking portfolio using:

- KPI cards
- Customer distribution by city
- Credit score distribution
- Average, minimum, and maximum credit scores
- Interactive page navigation

Accounts & Loans

Provides analysis of:

- Account balance by account type
- Accounts by account type
- Loan portfolio by credit score
- Loan interest rate distribution
- Card type distribution
- Card expiration by year

Transactions

Provides analysis of:

- Transaction value over time
- Transaction value by customer city
- Transaction value by account type
- Top merchants by transaction value
- Interactive transaction date filtering

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7. Key Insights

The analysis provides several important observations:

1. The banking dataset contains a large customer and account base suitable for analyzing customer and financial activity.
2. Customers are distributed across multiple credit score categories, providing visibility into different customer credit profiles.
3. Account balances vary across account types, indicating differences in how customers use banking products.
4. Loan exposure is distributed across different credit score categories, providing insight into the relationship between credit profiles and lending activity.

5. Transaction value changes over time, making monthly trend analysis useful for identifying periods of higher or lower activity.
6. Transaction activity differs across account types and customer cities.
7. Card expiration analysis can help identify future periods of higher card renewal activity.

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8. Business Recommendations

Based on the analysis:

- Monitor customer and account activity to identify high-value customer segments.
- Review loan exposure across different credit score categories.
- Compare account types based on both balances and transaction activity.
- Monitor monthly transaction trends for changes in customer behavior.
- Analyze high-value cities and merchants for potential business opportunities.
- Use card expiration trends to support proactive renewal planning.

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9. Conclusion

This project demonstrates an end-to-end **banking data analytics workflow** using MySQL and Power BI.

SQL was used to analyze customer, account, loan, transaction, card, merchant, and branch data, while Power BI was used to create an interactive dashboard.

The project demonstrates practical skills in:

SQL | Data Analysis | Power BI | DAX | Data Visualization | Business Insights

The final dashboard provides a concise view of banking performance across **customers, accounts, loans, and transactions**.

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10. Project Files

- Banking_Customer_Analytics.sql — SQL analysis

- Banking_Customer_Analytics_Dashboard.pbix — Power BI dashboard
- README.md — Project overview
- Banking_Customer_Analytics_Documentation.pdf — Detailed documentation